Lab 5: Quantum Computing Workshop

1 Superpostion

1.1 Calculations

We start with defining a basis state of $|0\rangle$:

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

To put the state in superpostion, the Hadamard gate needs to be applied.

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \xrightarrow{Hadamard} H|0\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{1}{\sqrt{2}} \left[\begin{pmatrix} 1 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right] = |+\rangle \tag{1}$$

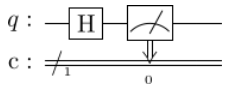
Comparting this to the general form of a qubit state in the given basis (where $|1\rangle = \binom{0}{1}$):

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \tag{2}$$

we can say that $\alpha = \beta = \frac{1}{\sqrt{2}}$ which means that $|\alpha|^2 = |\beta|^2 = \left|\frac{1}{\sqrt{2}}\right|^2 = .5$, or there is a 50/50 chance of being in ether state.

1.2 Code

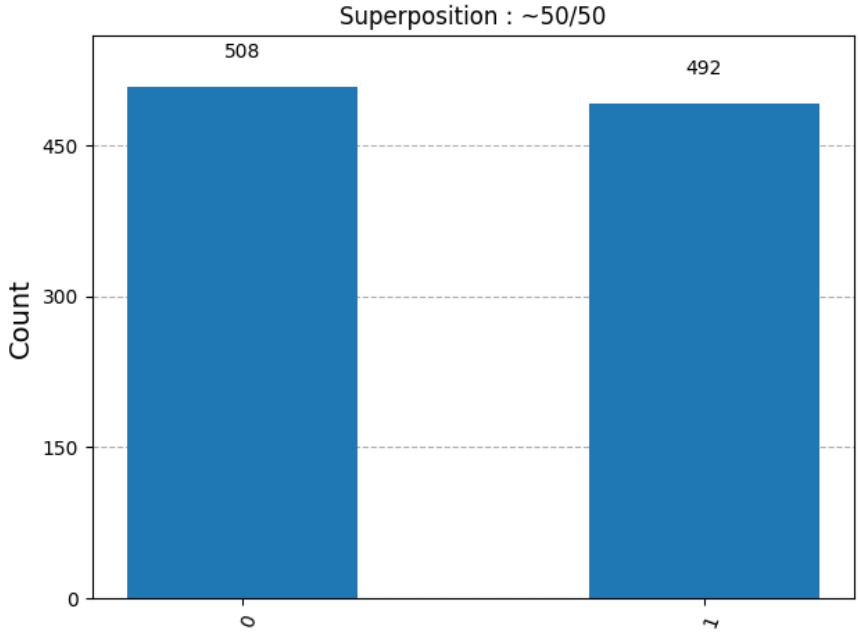
The circuit needed to implement the superpostion is shown below.



For reproducability the code that was used to simulate the quantum circuit is provided.

```
qc_super = QuantumCircuit(1,1)
qc_super.h(0)
qc_super.measure(0,0)
shots_super = 1024
job_super = simulator.run([qc_super], shots = shots_super)
plot_histogram(job_super.result().get_counts(0))
```

1.3 Results



The result of the simulation shows that the number of shots that were measured in the $|0\rangle$ state was 508 compared to the 492 shots measured in $|1\rangle$. The total number of shots ran was 1000, which means we had $\frac{508}{1000} \approx .51\%$ of the measurements result in $|0\rangle$ and $\frac{492}{1000} \approx .49\%$ of the measurements result in $|1\rangle$ which is within the 5% error margin expected.

2 Bell State

2.1 Calculations

We now have two qubits which means a new set of basis states are defined.

$$|00\rangle = \begin{pmatrix}1\\0\\0\\0\end{pmatrix}, \quad |01\rangle = \begin{pmatrix}0\\1\\0\\0\end{pmatrix}, \quad |10\rangle = \begin{pmatrix}0\\0\\1\\0\end{pmatrix}, \quad |11\rangle = \begin{pmatrix}0\\0\\0\\1\end{pmatrix}.$$

By starting with an initial state of $|00\rangle$ we can apply $H \otimes I$ then CNOT to create a Bell State.

$$(H \otimes I)\,|00\rangle = H\,|0\rangle \otimes I\,|0\rangle = |+\rangle \otimes |0\rangle = \frac{1}{\sqrt{2}}\left(|00\rangle + |10\rangle\right), \tag{3}$$

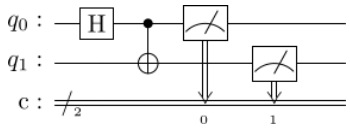
where the result of Eq. 1 was used to calculate $H\,|0\rangle$ and the main property of the Identity matrix was used to calculate $I\,|0\rangle$.

$$\text{CNOT}\,(H \otimes I)\,|00\rangle = \text{CNOT}\frac{1}{\sqrt{2}}\left(|00\rangle + |10\rangle\right) = \frac{1}{\sqrt{2}}\left(|0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes X\right)\left(|00\rangle + |10\rangle\right) = \frac{1}{\sqrt{2}}\left(|00\rangle + |11\rangle\right) \tag{4}$$

This result is similar to Part 1, but now we should have a 50/50 chance of our result being in $|00\rangle$ or $|11\rangle$ instead of $|0\rangle$ or $|1\rangle$.

2.2 Code

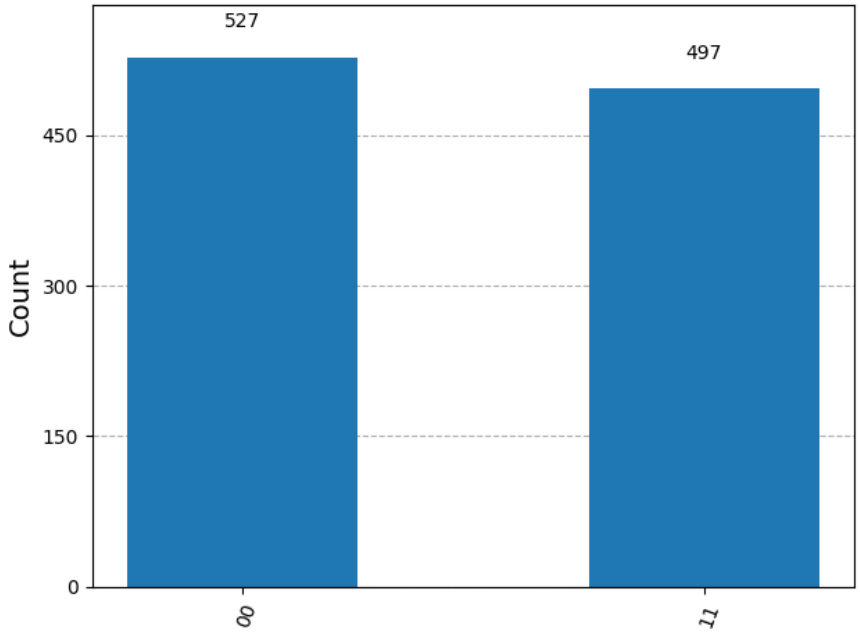
The circuit to implement the bell states is shown below.



For reproducability the code that was used to simulate the quantum circuit is provided.

```
qc_bell = QuantumCircuit(2,2)
qc_bell.h(0)
qc_bell.cx(0,1)
qc_bell.measure(0,0)
qc_bell.measure(1,1)
shots_bell = 1024
job_bell = simulator.run([qc_bell], shots = shots_bell)
plot_histogram(job_bell.result().get_counts(0))
```

2.3 Results



The result of the simulation shows that the number of shots that were measured in the $|00\rangle$ state was 527 compared to the 497 shots measured in $|11\rangle$. The total number of shots ran was 1024, which means we had $\frac{527}{1024} \approx .51\%$ of the measurements result in $|00\rangle$ and $\frac{497}{1024} \approx .49\%$ of the measurements result in $|11\rangle$ which is within the 5% error margin expected. These percentages add up to 100%, which is expected as the only two results shown in the histogram are $|00\rangle$ and $|11\rangle$.

3 Modified Bell

3.1 Calculations

Using the same basis states and circuit from part 2, we modify the circuit and add a X (bit-flip) gate to qubit 1 (the second qubit) before the CNOT. This means we are now starting with an initial state of $|00\rangle \xrightarrow{I\otimes X} |01\rangle$, then apply $H \otimes I$ then CNOT to create a Modified Bell State.

$$(H \otimes I)\,|01\rangle = H\,|0\rangle \otimes I\,|1\rangle = |+\rangle \otimes |1\rangle = \frac{1}{\sqrt{2}}\left(|01\rangle + |11\rangle\right), \tag{5}$$

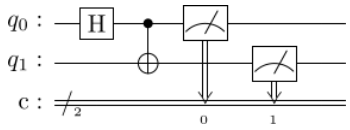
where the result of Eq. 1 was used to calculate $H\,|0\rangle$ and the main property of the Identity matrix was used to calculate $I\,|1\rangle$. As the modification done to the circuit only effects the second qubit, this means the result should be simmilar to Part 2, whith

$$\text{CNOT}\,(H \otimes I)\,|01\rangle = \text{CNOT}\frac{1}{\sqrt{2}}\left(|01\rangle + |11\rangle\right) = \frac{1}{\sqrt{2}}\left(|0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes X\right)\left(|01\rangle + |11\rangle\right) = \frac{1}{\sqrt{2}}\left(|01\rangle + |10\rangle\right) \tag{6}$$

This result is similar to Part 1, but now we should have a 50/50 chance of our result being in $|00\rangle$ or $|11\rangle$ instead of $|0\rangle$ or $|1\rangle$.

3.2 Code

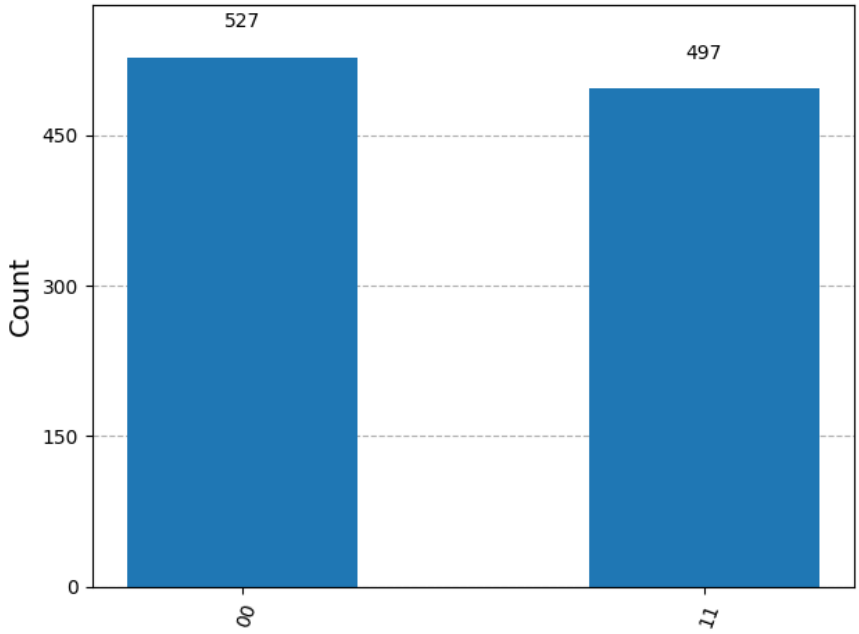
The circuit to implement the bell states is shown below.



For reproducability the code that was used to simulate the quantum circuit is provided.

```
qc_bell = QuantumCircuit(2,2)
qc_bell.h(0)
qc_bell.cx(0,1)
qc_bell.measure(0,0)
qc_bell.measure(1,1)
shots_bell = 1024
job_bell = simulator.run([qc_bell], shots = shots_bell)
plot_histogram(job_bell.result().get_counts(0))
```

3.3 Results



The result of the simulation shows that the number of shots that were measured in the $|00\rangle$ state was 527 compared to the 497 shots measured in $|11\rangle$. The total number of shots ran was 1024, which means we had $\frac{527}{1024} \approx .51\%$ of the measurements result in $|00\rangle$ and $\frac{497}{1024} \approx .49\%$ of the measurements result in $|11\rangle$ which is within the 5% error margin expected. These percentages add up to 100%, which is expected as the only two results shown in the histogram are $|00\rangle$ and $|11\rangle$.